Airframe Design Theory Exploration: Part I

In this assignment, you will explore the basic concepts of airframe design, including wing geometry, forces and moments, and the use of VortexLattice.jl to analyze these concepts. In order to accomplish this, you will also need to read about the fundamentals of flight vehicle design and drag. You might not have done a lot of text book reading in the past, but learning to learn from textbooks is a valuable skill that will serve you well in your career. It is critical to developing your ability to learn new things on your own, and to develop a deep understanding of the subject matter.

Relevant Reading

- Flight Vehicle Design Book
  - Chapter 2: Fundamentals (13 pages)
  - Chapter 3: Drag (28 pages)

Problem 1 Vocabulary

Explain the following terms; making sure to use sufficient detail, including any math or helpful figures. In some cases, these terms are simple one sentence definitions, in others, you should include several paragraphs to explain them fully.

Wing Geometry Terms

- Wing Area
- Chord
  - Mean Geometric Chord
  - Mean Aerodynamic Chord
- Taper Ratio
- Span
- Aspect Ratio
- Sweep
- Dihedral
- Twist
- Washout

Forces and Moments

- Lift
- Drag
  - Induced Drag
  - Parasitic Drag
    - Skin Friction Drag
    - Pressure Drag
  - Compressibility Drag
- Pitching Moment
- Lift and Drag Polars
  - Angle of Attack
  - Zero Lift angle of attack
  - Lift Curve Slope
  - Stall

Problem 2 Exploration

Complete the following exploration. Present your exploration in a way that is easy to read and understand. Use figures, tables, and equations as needed to clearly communicate your findings. You'll notice that in these earlier assignments have more sub-questions in the exploration portion of the assignment to help get you started. As the semester progresses, the exploration section will become more open-ended, and you will be expected to provide the digging questions yourself.

2.a Prerequisites

i. Install VortexLattice.jl

ii. Run and annotate the Getting Started Guide. Your annotations should describe the purpose of each block of code, what is going in, and what is coming out.

- Note: The documentation doesn't explain this well: the X direction is free-stream direction, the Y direction is the span-wise direction (out the starboard wing), and the Z direction is the vertical direction (up).

2.b Investigate the Force and Moment Polars.

i. Create the following plots using the example wing from the previous problem:

(a) Lift vs Angle of Attack

(b) Induced Drag (near and far field) vs Angle of Attack

(c) Moment vs Angle of Attack

(d) Lift vs Drag

(e) Lift/Drag vs Angle of Attack

ii. Identify the lift curve slope in your lift vs angle of attack plot, and compare it to the lift curve slope from thin airfoil theory (2π).

(a) What is the error between the two values?

(b) What is the cause of this error? Is there theoretical reason why these should be different? How could you decrease this error?

iii. Consider your lift vs angle of attack plot, is this what you expect it to look like? Why? If not, what things are missing? Why would they be missing? How does it compare to experimental/real-word data?

iv. What other trends do you see in the plots? What do you think is causing these trends? What do they tell you about the wing? What do they tell you about the model?